

Methods: detection of coordinated calcium events

Input. A *coordinated event* is a group of calcium-event onsets in several regions of interest (ROIs, one cell each) within about a second; a detector reports each as a *call*, an interval of time. The input is the imaging pipeline's per-ROI event table: each event's onset, its half-rise time (t_{50}), on a 0.1 s frame grid. The pipeline exports two event sets, *fast* and *slow*, assigned upstream; a third, *combined*, stream is the time-sorted union of each ROI's fast and slow onsets, near-coincident pairs kept. *Windows* are each recording's baseline and treatment periods. The data are 66 recordings (2,109 ROIs, 36 mice) whose first treatment was tetrodotoxin (TTX) or senktide, in four groups: diestrus females (DI, 17 recordings), ovariectomized females (OVX, 17), males (MALE, 13) and orchidectomized males (ORX, 19). Group is nested in imaging day.

Event floor. A call needs at least the *event floor* of ROIs co-active within 2 s: the smallest count chance reaches at most once per hour, and never fewer than 3 ROIs. Chance comes from 1,000 draws that shift each ROI's onset train as a whole by its own offset, uniform in $[-20\text{ s}, 20\text{ s}]$, keeping each ROI's rate and intervals (after Pipa et al., 2007, but by up to 20 s and dropping rather than wrapping onsets pushed past the window's edge). The null keeps coordinated onsets, so coordination-rich windows get a higher, conservative floor. Each window of each stream, and each simulated recording, has its own floor; treatment windows are also scored under their baseline floor.

Parameters for simulation. Detectors were tuned and scored on simulated recordings with known (*planted*) coordinated events, measured on baseline windows, groups pooled. *Spacing*: coordinated events were located, without the detectors compared here, as runs of the co-active count at or above the floor (peaks under 2 s apart merged). Median gaps between neighbors were 41.3 s (fast), 25.1 s (slow) and 24.8 s (combined), at 9.7, 22.0 and 25.3 events per hour of baseline (gaps are skewed). *Participation*: median participants per moment with at least 4 ROIs co-active within 1 s, over median ROI count (fast 0.20, slow 0.38, combined 0.25) set the middle of three planted levels; the outer two were chosen. *Background*: quiet and busy per-ROI event rates are the 25th and 75th percentiles across recordings, coordinated onsets removed, varied across ROIs and over time by Gamma factors fitted to the data. *Timing spread*: the standard deviation (SD) of onsets within an event (fast 0.105 s, slow 0.131 s, combined 0.150 s) came from the half-width of the cross-ROI onset correlogram (Perkel et al., 1967), calibrated on simulations of known spread.

Simulated recordings. Each has 32 ROIs (the measured median) and lasts 45 min. It plants the stream's measured event rate over 45 min (fast 7, slow 17, combined 19) over the three levels, laid end to end at gaps resampled from the pooled gaps and placed at a random time, plus six *decoys* (bursts like middle-level events, not scored), on a quiet or busy background. Each seed also yields a *no-coordination recording* (quiet, nothing planted) and an *elevated-rate recording* (a 300 s stretch at the 99th percentile of background rate). A variant spaced like ORX checks the rankings. Separate seeds serve *selection*, *held-out* confirmation and *final* scoring, doubled on fast.

Scoring. A call matches a planted event lying in its span or within 2.5 s of it, one to one. A planted event under its recording's floor, and any call matched to it, leave the score. F1, the harmonic mean of precision and recall, is pooled per background and averaged over the two; calls on decoys count against precision. Calls holding two or more scored events are counted as merges. Every CoactDetect setting must meet budgets set from the measured rates of its shipped settings with headroom (1.6 $\times$, never below 1): calls per hour on the no-coordination recording (fast 7, slow 1, combined 10), 1 call per minute in the elevated stretch, and a quiet-to-busy precision difference of at most 0.10.

CoactDetect. $S(t)$ counts the ROIs with an onset in $(t - w, t]$, w being the count interval. Its mean and variance under a null that circularly shifts each ROI's onsets within a surrounding *context*, optionally excluding a guard band around t , are computed exactly; t is called when S reaches the floor and $z = (S - \text{mean})/\text{SD}$ passes a cutoff α stated as a normal tail, a z threshold. Calls closer than a merge gap are joined. The design resembles cell-averaging constant-false-alarm-rate radar detection (Finn & Johnson, 1968) and excess-coincidence tests against a shifted null (Amarasingham et al., 2012), reached independently. Settings were searched in this sliding form (fast ships fixed 2 s bins) by coordinate search from the shipped settings: α down to 6×10^{-16} ($z \approx 8$), contexts 20–120 s, guard at most a quarter of the context. The floor is always computed at 2 s, so its once-per-hour guarantee holds only at $w = 2$ s. A candidate was adoptable if its held-out F1 gain had a 95% bootstrap interval above zero, its best values lay inside the tested range, and it met every budget; a best value at a cap, or one that switches a setting off, was reported, not adopted.

Chorus. Chorus, a learned detector, reads a binary ROI $\times$ frame onset raster. A four-layer dilated convolutional stack (Yu & Koltun, 2016) shared by all ROIs filters each ROI; each channel is standardized over the whole input and passed through a sigmoid as a per-ROI vote. Votes are pooled across ROIs (mean, SD, mean of the top four), so any number of ROIs in any order is accepted (Qi et al., 2017; Zaheer et al., 2017), and a dilated head scores each frame; frames over a threshold chosen on training-side validation recordings become calls, runs within 2 s merged. Standardization spans 409.6 s excerpts in training and the whole window at inference. A participation variant sums the votes into a bounded count, takes the floor as input and learns which ROIs took part. Training was per stream (Adam, Kingma & Ba, 2015; learning rate 0.01; 900 steps; class-weighted binary cross-entropy), with extra events planted at the floor and one ROI either side, sub-floor events labelled negative. Of five initializations, the best on selection recordings within the no-coordination budget was kept (the best overall if none met it).

Benchmark. Each detector was compared with CoactDetect on the final seeds as a paired per-seed F1 difference (each seed's quiet and busy recordings pooled) with a 95% bootstrap interval over seeds (2,000 resamples; Efron, 1979).

Output. Every call is measured alike, whatever its detector. Onsets within ± 1 s (fast) or ± 5 s (slow, combined) of the call's center, widened to its span, are split where consecutive onsets are over 0.5 s (fast) or 2.5 s (slow, combined) apart; the group with most distinct ROIs is the event's *core*. These lengths were chosen, not measured. The event time is the core's first onset; *width* is the core's first-to-last onset span; *amplitude* is core ROIs divided by width floored at one frame (ROIs s^{-1}), undefined for one ROI. Amplitude measures packing, not fluorescence, is dominated by width, and has a provisional name. On simulated data the core's ROI count never under-counts planted participants but over-counts, most on combined (median +2–3 ROIs); width and amplitude are unchecked.

References

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